Why Are HPC Monitoring Signals Hard to Forecast?

Intrinsic Predictability Across Nine Telemetry Families of a Tier-0 Supercomputer

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Abstract

Operational telemetry from high-performance computing systems resists forecasting to a degree that is widely reported anecdotally and rarely explained. The usual response is to increase model capacity. We argue instead that the difficulty is largely *intrinsic to the signals*, differs systematically between signal families, and can be quantified without fitting any forecaster.

Using the public M100 ExaData release from CINECA's Marconi100 Tier-0 supercomputer, we characterise telemetry from nine heterogeneous plugin families — out-of-band sensors, operating-system counters, scheduler aggregates, alarm states, facility power, cooling-plant control registers, and exogenous weather — collected from the same machine over the same 31 months, so that signal type varies while system, workload, and period are held fixed. For each series we compute model-free predictability measures spanning entropy, periodicity, long memory, non-stationarity, and quantisation, and separately measure the forecast skill achieved by a tiered suite ranging from naive baselines to gradient-boosted trees under a strictly causal rolling-origin protocol.

We report three findings. First, the monitoring infrastructure is itself non-stationary: of 573 distinct (plugin, metric) pairs observed across the campaign, only 104 are present in every month, and those are exactly the out-of-band sensor set. Second, **[predictability landscape result]**. Third, **[linking result: out-of-sample R^2 under leave-one-family-out cross-validation]**.

[Implications paragraph.]

1 Introduction

[To be written.] Framing points to develop:

- Operators increasingly collect holistic telemetry, and increasingly want to *predict* it — for thermal safety, power capping, and proactive maintenance — not merely to detect anomalies after the fact.
- Reported forecast accuracy on this class of signal is poor relative to standard time-series benchmarks, and the literature treats this as a modelling deficiency.
- We invert the question: rather than asking which model does best, we ask how much predictability is present at all, measured independently of any model.
- Contributions: (i) a quantitative census of signal heterogeneity in a Tier-0 monitoring campaign, including the instability of the monitoring system itself; (ii) a model-free predictability profile per signal family; (iii) a causal, scale-free forecast benchmark on the same series; (iv) a test of whether the first predicts the second, validated by holding out entire signal families.

Table 1: Telemetry families and their generating mechanisms. Nominal cadences are as documented by the dataset; measured cadences differ substantially for some families (§5.1).

Family	Measures	Nominal cadence	Generating mechanism
IPMI	BMC sensors: temperature, power, fans	20 s	thermal diffusion, quantised
Ganglia	OS and GPU counters	20–300 s, irregular	software instrumentation
SLURM	cluster scheduler aggregates	5–10 s	user submission behaviour
Nagios	service and hardware alarm states	15 min	discrete state machine
Job table	per-job submission records	event-driven	not a time series (OI-8)
Schneider	liquid-cooling PLC registers	20 s	closed-loop control
Logics	room power, PUE, chillers	—	facility aggregation
Vertiv	air-handling units	10 s	closed-loop control
Weather	outdoor conditions	10 min	exogenous

2 Background

2.1 Marconi100 and the ExaData campaign

[Machine description and campaign scope.]

2.2 The nine plugin families

The value of this dataset for the present question is that it holds the system fixed while varying the signal. Table 1 summarises the families and the mechanism that generates each.

3 Related work

[To be written from the screening table in `literature/screening.csv`.] Every claim in this section must be verified against the source PDF before submission; none has been, and this is tracked as OI-10.

4 Method

4.1 Campaign census

[Describe the inventory and per-series census.]

4.2 The entity panel

Node-level series of one metric are highly redundant — nodes share racks, inlet air, cooling zones, and workload — and are therefore not independent samples (OI-9). Analysis runs on a fixed panel of 96 entities whose membership is a stable hash of the entity label under the project seed. Because the hash sees only the label, panel membership is identical across every metric and every month, so cross-metric comparisons are made on the same physical machines rather than on a fresh random slice.

4.3 A-priori signal taxonomy

Labels are assigned from the plugin and metric *name* alone, before any predictability is measured, so that a finding of between-family differences cannot be a relabelling of the result it is meant to explain.

4.4 Model-free predictability measures

[Table of measures and what each probes.]

4.5 Forecast protocol

Skill is measured under rolling-origin evaluation with strict causality asserted per split. Errors are scaled by MASE with a denominator computed from the training window only; scaling by test-window statistics is the most common silent leak in forecasting benchmarks and would compress exactly the between-tier differences this study reads. The seasonal period is fixed per series *a priori* from its own cadence, with a non-seasonal robustness check (OI-5).

MASE is undefined where the in-sample seasonal-naive error vanishes. This is not an edge case here: a substantial share of control-plane series are constant (§5.1), and near-perfectly periodic windows yield denominators at the level of floating-point noise. Such series are reported as undefined rather than assigned a spuriously large score.

4.6 Linking analysis

[Describe leave-one-family-out CV, baseline, ablations.]

5 Results

5.1 The monitoring system is not stationary

Across the 31-month campaign we catalogue 10504 metric partitions comprising 573 distinct (plugin, metric) pairs. Only 104 — all of them IPMI — are present in every month. Ganglia's metric count runs 0, 33, 73, 49, 157, 177 and is not monotonic; SLURM's runs 20, 2, 0, 2, 36; Schneider first appears in 21-03 and Vertiv and Weather in 21-04.

Any claim compared across months therefore confounds signal behaviour with changes to the instrument, and we restrict cross-month statements accordingly (OI-2).

[Figure: coverage over time.]

5.2 Signal heterogeneity

[Census results: cadence, cadence irregularity, coverage, quantisation, constant fraction, per family.]

5.3 The predictability landscape

[Feature distributions per family.]

5.4 Forecast skill

[Skill by tier, horizon, and family; the saturation test.]

5.5 Does predictability explain skill?

[Leave-one-family-out R^2, importances, ablations.]

6 Threats to validity

The full register is maintained in `OPEN_ISSUES.md` and each item is cited from the text above. The load-bearing ones are:

oi-1 Ganglia's sampling irregularity confounds instrumentation with signal physics. Sampling regularity is therefore carried as a covariate and reported separately.

oi-2 The monitoring system changes over the campaign.

oi-6 No neural tier has been run; no claim of the form "not even deep models help" is licensed.

oi-7 The error floor is an estimate with a known bias direction, not an information-theoretic bound.

oi-11 Permutation entropy is biased downward by quantisation, and quantisation varies by family, so entropy is interpreted only alongside the measured quantisation step.

7 Conclusion

[To be written.]

Data and code availability

All code, the versioned literature query set, the screening table, and every derived table are at `https://github.com/MSKazemi/hpc-signal-forecastability`. The underlying measurements are the public M100 ExaData release [Borghesi et al., 2023]. Results reproduce from the committed derived tables without downloading raw data.

References

Andrea Borghesi, Carmine Di Santi, Martin Molan, Mohsen Seyedkazemi Ardebili, Alessio Mauri, Massimiliano Guarrasi, Daniela Galetti, Mirko Cestari, Francesco Barchi, Luca Benini, Francesco Beneventi, and Andrea Bartolini. M100 ExaData: a data collection campaign on the CINECA's Marconi100 Tier-0 supercomputer. *Scientific Data*, 10(1):288, 2023. doi: 10.1038/s41597-023-02174-3.